HW 4 Due 2/9

1) 3.4 29

$\{\{1,2,3\},\{4\},\{5\},\{6\}\}$

$[1]=[2]=[3]=\{1,2,3\}$

$[4]=\{4\}$

$[5]=\{5\}$

$[6]=\{6\}$

$R=\{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6),(1,2),(2,1),(1,3),(3,1),(2,3),(3,2)\}$

2) 3.4 32

✓ Reflexive: $f R f$ must be true. $f(0)=f(0)$, so fRf, the relation is reflexive.

✓ Symmetric: If fRg, gRf. fRg means $f(0)=g(0)$. This implies $g(0)=f(0)$, so fRg does mean gRf.

✓ Transitive: If fRg and gRh, fRh as well. fRg means $f(0)=g(0)$. gRh means $g(0)=h(0)$. Thus, $f(0)=g(0)=h(0)$, $f(0)=h(0)$, so fRh. R is an equivalence relation.

$[f]=\{g:\mathbb{R}\to\mathbb{R} \mid g(0)=0\}$

3) 3.4 34

✓ a) Reflexive: $(a,b)R(a,b)$ $a+b=b+a$. Since this is true, R is reflexive, because $(a,b)R(a,b)$.

✓ Symmetric: If $(a,b)R(c,d)$, $(c,d)R(a,b)$. $(a,b)R(c,d)$ means $a+d=b+c$. $(c,d)R(a,b)$ means $c+b=d+a$. This is equivalent to the last equation, so it is true, so when $(a,b)R(c,d)$, $(c,d)R(a,b)$, the relation is symmetric.

✓ Transitive: Let $(a,b)R(c,d)$ and $(c,d)R(e,f)$. This means $a+d=b+c$, and $c+f=d+e$. For $(a,b)R(e,f)$, it must be true that $a+f=b+e$. Manipulating $c+f=d+e$, we get $f=d+e-c$. substitute: $a+d+e-c=b+e$. $a+d=b+c$. We saw this was true with $(a,b)R(c,d)$, so $(a,b)R(e,f)$. R is transitive.

R is reflexive, symmetric, and transitive, so R is an equivalence relation on $X\times X$.

34) b) $[1,2] = \{(3,4)\ldots\}$ $[1,1] = \{(2,2)\ldots\}$

$[1,3] = \{(3,5),\ldots\}$

$[1,4] = \{(3,6)\ldots\}$

$[1,5] = \{(3,7)\ldots\}$

$[1,6] = \{(3,8)\ldots\}$

$[1,7] = \{(3,9)\ldots\}$

$[1,8] = \{(3,10),\ldots\}$

$[1,9] = \{(2,10)\ldots\}$

$[1,10] = \{(10,1)\}$

$[2,1] = \{(4,3)\ldots\}$

$[3,1] = \{(5,3),\ldots\}$

$[4,1] = \{(6,3)\ldots\}$

$[5,1] = \{(7,3),\ldots\}$

$[6,1] = \{(8,3)\ldots\}$

$[7,1] = \{(9,3)\ldots\}$

$[8,1] = \{(10,3)\ldots\}$

$[9,1] = \{(10,2)\ldots\}$

$[10,1] = \{(1,10)\}$

4) 41) xRy iff $f(x) = f(y)$

✓ Reflexive: $f(x) = f(x)$ so xRx for all x in Y, so the relation is reflexive.

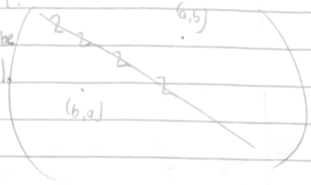
✓ Symmetric: If xRy, $f(x) = f(y)$. This means $f(y) = f(x)$, so yRx too. The relation is symmetric.

✓ Transitive: If xRy, $f(x) = f(y)$. If yRz, $f(y) = f(z)$. This means $f(x) = f(y) = f(z)$. This also means $f(x) = f(z)$, so xRz. The relation is transitive.

Thus, R is an equivalence relation on Y.

7) 87) To be antisymmetric, if (a,b) is in the relation, (b,a) cannot be, since $b \neq a$. Either one can be in the relation, the other can, or neither can — 3 choices. The diagonal can or cannot be in the relation. But according to symmetry, if aRb is in a relation, bRa must be too. This leaves just the diagonal, which can either be in the relation or not.

If (a,b) is in the relation and (b,a) must be (symmetric) - then $a = b$ (antisymmetry). This leaves the diagonal only.



2^n

8) 88) R, S, A

Since the relation is reflexive, the diagonal must be included. As per 87, the diagonal is the whole relation. If anything off the diagonal is included, the opposite would also have to be to fulfill the symmetric property, but this would violate antisymmetry since $a \neq b$. Overall, we are left with $n = 1$ relation.



5) ~~27 28 30 31 33~~

27) $\underline{\quad}\ \underline{\quad} = 6 \cdot 1 = \boxed{6}$

28) $\underline{\quad}\ \underline{\quad} + \underline{\quad}\ \underline{\quad} = 6 \cdot 2 = \boxed{8}$

1+6 5+6
2+5 6+5
3+4
4+3
5+2
6+1

30) $\underline{\quad}\ \underline{2} + \underline{2}\ \underline{\quad}$

$5 \cdot 1 + 1 \cdot 5 = \boxed{10}$

31) can add answer from 30 (exactly one die showing 2) to outcomes where both die are 2. $\frac{2}{1}\frac{2}{1}$

$10 + 1^2 = \boxed{11}$

33) even+even odd+odd

$\underline{\quad}\ \underline{\quad} + \underline{\quad}\ \underline{\quad}$

$3^2 + 3^2 = \boxed{18}$

6) 85) Symmetric relations: 2 choices in upper triangle, either included or not, the lower triangle matching symmetric pair will only be included if the other is, so do not need to consider. Only one choice for diagonal, need to be included for reflexive relation.



$2^{\frac{n^2-n}{2}} \cdot 1^n = 2^{\frac{n^2-n}{2}}$